

Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flask development server is single-threaded	Cannot handle many concurrent users in production	Medium
2	Model trained on a single reference year of data	No trend-based or time-series predictions possible	Medium
3	No user authentication or history storage	Users cannot save or revisit past predictions	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local user via Flask dev server	Deploy with Gunicorn/Nginx or a managed platform (Render, Heroku, AWS) for concurrent access
2	Data Storage	Static CSV file, no database	Migrate to a lightweight database (SQLite/PostgreSQL) to support multi-year data and history

3	Performance	Model loads once at startup; fast local inference	Cache predictions and add async request handling for higher throughput
4	Security	No authentication, open access	Add user authentications if history/personalization features are introduced.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment (Render/Heroku) with public URL	1 month post-submission	Makes the tool publicly accessible beyond localhost
Phase 2	Multi-year HDI dataset support with trend visualization	2 months	Enables year-over-year HDI trend predictions
Phase 3	Add Inequality-adjusted HDI and Gender Development Index	3 months	Provides a more nuanced development picture
Phase 4	Country dropdown with auto-filled known indicators	3 months	Improves usability for non-expert users